

Q1. What is Software Quality Assurance (SQA)?

Software Quality Assurance is a way to make sure that software is made the way. It follows rules and standards that are already set. The goal of Software Quality Assurance is to stop problems from happening in the place. This means making the process of developing software. Software Quality Assurance does not just look for mistakes after the software is made. It tries to improve the process of making software so that Software Quality Assurance can prevent mistakes from happening.

Software quality is important because:

1. Customer Satisfaction
2. Reduces Cost
3. Improves Reliability
4. Ensures Security
5. Saves Time
6. Enhances Business Reputation

Q2. Differentiate between the following:

A. QA vs QC

QA (Quality Assurance)	QC (Quality Control)
Process-oriented	Product-oriented
Prevents defects	Detects defects
Focuses on improving processes	Focuses on testing products
Done throughout development	Done after development activities
Preventive approach	Corrective approach

B. QC vs Testing

Quality Control	Testing
Overall quality checking process	Activity to identify software defects
Includes reviews, inspections, testing	Only executes test cases
Broader concept	Subset of QC
Ensures product quality	Finds bugs and verifies functionality

C. QA vs Testing

Quality Assurance	Software Testing
Focuses on preventing defects	Focuses on finding defects
Covers the entire SDLC	Performed during testing phase
Improves development process	Verifies software functionality
Process-oriented	Product-oriented

Q3. How is AI changing Software Testing?

Artificial Intelligence is transforming software testing by making it faster, smarter, and more efficient.

1. Automated Test Case Generation

AI automatically generates test cases based on application behavior.

2. Self-Healing Test Scripts

AI updates broken test scripts automatically when UI changes occur.

3. Intelligent Bug Prediction

Machine learning predicts high-risk areas where defects are likely.

4. Visual Testing

AI compares screenshots and detects visual differences automatically.

5. Test Prioritization

AI selects the most important test cases based on previous failures.

6. Faster Regression Testing

AI reduces testing time by executing only relevant test cases.

7. Defect Analysis

AI groups similar bugs and identifies root causes quickly.

Q4. Explain the responsibilities of:

A. QA Engineer

Responsibilities

1. Create quality standards
2. Prepare QA plans
3. Review requirements
4. Improve development processes
5. Conduct audits

Daily Activities

1. Attend project meetings
2. Review requirements
3. Prepare QA documentation
4. Monitor testing progress
5. Report quality issues

B. Software Test Engineer

Responsibilities

1. Design test cases
2. Execute manual tests
3. Perform automated testing
4. Report software defects
5. Verify bug fixes

Daily Activities

1. Analyze requirements
2. Write test cases
3. Execute tests
4. Log bugs
5. Retest fixed issues

C. AI Quality Engineer

Responsibilities

1. Test AI/ML models

2. Validate AI predictions
3. Check model accuracy
4. Detect AI bias
5. Evaluate datasets

Daily Activities

1. Validate datasets
2. Test AI outputs
3. Analyze prediction accuracy
4. Perform model evaluation
5. Monitor AI systems

Q5. Draw the career roadmap of a QA professional.	

QA PROFESSIONAL CAREER ROADMAP

